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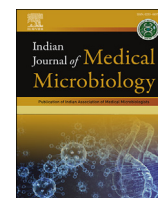


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Original Research Article

Burden of biocide resistance among multidrug-resistant bacteria isolated from various clinical specimens in a tertiary care hospital

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ABSTRACT

Background: Most studies on biocide resistance and its genetic determinants arise from environmental or food-borne microbial isolates and only a few from clinically relevant isolates.

Objectives: This study determines the proportion of biocide resistance against five commonly used biocides and detects biocide resistance genes among MDR bacterial isolates using PCR.

Methods: Consecutive MDR isolates (n = 180) were included (30 each of *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Staphylococcus aureus*, and *Enterococcus* species) from clinical specimens of various inpatient units at JIPMER. The isolates were challenged at 0.5, 1 and 2 Macfarland (McF) inoculum with discrete dilutions of disinfectants. The minimum bactericidal concentrations (MBCs) for 70% Ethanol, 1.5% Cresol, 2% Glutaraldehyde, 1% Cetrimide, and 1% Chlorhexidine were determined for the isolates using ATCC reference strains as controls. PCR was performed targeting *qac* A/B, G; *smr*; and *nfx* B genes.

Results: For all biocides, MDR isolates had MBCs less than the maximum MBCs of ATCC strains. For MDR *K. pneumoniae*, *A. baumannii*, and *P. aeruginosa*, the highest MBCs of chlorhexidine and cetrimide were ≥ 75 and ≥ 150 $\mu\text{g/ml}$ respectively at 0.5 McF inoculum; whereas these organisms grew at higher inoculum (2McF) even at commercially recommended biocidal concentration (1%) corresponding to 750 and 1500 $\mu\text{g/ml}$ of chlorhexidine and cetrimide respectively. Meanwhile, the highest MBCs of MDR *E. coli* were 75 and 150 $\mu\text{g/ml}$ for chlorhexidine and cetrimide respectively. Interestingly, the Gram-positive cocci survived the action of up to 35% ethanol. The *nfxB* and *qacG* genes were detected in 87% and 6.67% of MDR *P. aeruginosa* isolates respectively with no biocide resistance genes detected among the other organisms.

Conclusions: Biocide dilutions challenged with higher inoculum indicated a narrow margin of effectiveness for certain biocides. Although a significant proportion of clinical MDR isolates of *P. aeruginosa* harbored biocide resistance genes, this finding had no phenotypic correlation.

1. Introduction

The microbial co-resistance to antibiotics and biocides in hospital settings has received considerable attention in the literature [1–3]. There has been significant debate as to whether biocide resistance and antibiotic resistance are in any way linked – with evidence both for and against this phenomenon [4,5]. Differences between these two groups of agents are widely documented - with biocides having a much broader spectrum of activity and multiple non-specific sites of action as compared to specific antibiotic targets; biocides being used at concentrations much higher than their organism-specific minimum bactericidal concentrations (MBC's) compared to antibiotics; and mode of

application (such as surface-based for biocides vs systemic administration for antibiotics) [6].

However, similarities exist between the actions of antibiotics and biocides – exemplified by the self-promoted entry of cationic antibiotics and biocides into Gram-negative bacteria [3]. These similarities run deeper with the potential overlap between resistance mechanisms to antibiotics and biocides. Several MDR gene families (mostly encoding active efflux mechanisms) are implicated in biocide and antibiotic resistance (cross-resistance) [4,7–9]. These include the - SMR (Small Multidrug Resistance), MFS (Major Facilitator Superfamily), RND (Resistance/Nodulation/Cell Differentiation superfamily) and MATE (Multidrug And Toxic compound Extrusion) gene families [3,9,10]. Also, akin to antibiotic resistance -

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biocide resistance may be either intrinsic (mostly related to membrane/cell wall structure, efflux pumps, or formation of endospores and biofilms) or acquired (via genetic mutations or horizontal transfer of mobile genetic elements containing biocide resistance genes) [10].

Unlike antibiotic resistance, the study of biocide resistance poses several challenges, such as the lack of scientific consensus as to the exact definition of biocide resistance (reflected in the lack of susceptibility breakpoints), uncertainty regarding the concentrations of biocides and exposure times included in testing as well as inadequate knowledge regarding mechanisms of action of and resistance to biocides [2,6,7]. The most common method for assessment of biocide efficacy/resistance is via measurement of minimum inhibitory concentrations (MIC) by macro-/micro broth dilution or agar dilution [6,11]. However, the desired endpoint for most biocide applications involves the inactivation/killing of micro-organisms, not inhibition of their growth – hence minimum bactericidal/biocidal concentrations (MBCs) are sometimes considered as a more appropriate metric for biocidal efficacy [6,12].

2. Materials and methods

2.1. Source and selection of multi-drug resistant (MDR) bacterial isolates and control strains

A total of 180 consecutive MDR bacterial isolates (defined as isolates that are not susceptible to at least one agent in at least three antimicrobial classes) were collected (i.e. 30 isolates each for *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Staphylococcus aureus*, and *Enterococcus* species; as identified by Matrix Assisted Laser Desorption Ionization-Time of Flight Mass Spectrometry/MALDI-TOF MS, bioMérieux, Marcy-étoile, France). The clinical isolates were collected for a period between August 2021 to December 2022 from clinical specimens of patients admitted into various ICUs and other inpatient units at JIPMER. The collected isolates were stocked at 2–4 °C in nutrient agar containing Bijou bottles as well as at –20 °C in Luria Bertani (LB) broth containing 20% glycerol. American Type Culture Collection (ATCC) *Escherichia coli* 25922, ATCC *Pseudomonas aeruginosa* 27853, ATCC *Staphylococcus aureus* 25923 and ATCC *Enterococcus faecalis* 29212 were selected as the control strains.

2.2. Determining the minimum bactericidal concentration (MBC) of biocides by tube dilution method

Inoculum preparation was prepared in Cation Adjusted Mueller Hinton Broth (CAMHB) by incubation overnight at 37 °C, followed by checking for the presence of visible turbidity in the growth medium and adjustment of the same to 0.5 McF, 1 McF, or 2 McF using a densitometer (bioMérieux, France) depending on the biocide tested. Serial twofold dilutions of the biocides (70% Ethyl Alcohol, 1.5% Cresol, 2% Glutaraldehyde, 1% Cetrimide, and 1% Chlorhexidine) were prepared and mixed with the prepared inocula as illustrated in Table 1 (Supplementary file). Tenfold dilutions of

the biocides 1% Cetrimide and 1% Chlorhexidine were also prepared for testing their action at lower concentrations and against higher inoculum (method of dilution shown in Table 2-Supplementary file).

2.3. Biocide resistance gene detection by conventional polymerase chain reaction

The stocked bacterial isolates were revived by inoculation into Brain Heart Infusion (BHI) broth followed by overnight incubation. DNA extraction was performed from the revived bacterial isolates by following the boiling lysis method for the Gram-negative bacilli and using the QIAamp® DNA Mini Kit (QIAGEN) by following the manufacturer's instructions for the Gram-positive cocci. The extracted DNA elute was stored at –20 °C till use.

Conventional PCR was performed on the extracted nucleic acid after standardization of reaction conditions to detect the genes responsible for biocide resistance, namely the *nfxB* and *qacG* genes in Gram-negative bacilli; and *qacA/B*, and *smr* in Gram-positive cocci. The primer sets and PCR reaction conditions used for the amplification process are listed in Tables 3 and 4 respectively.

3. Results

3.1. MBC determination for the MDR bacterial isolates

The MBCs of the various biocides were obtained for the 180 isolates included in the study (with 30 strains belonging to each of the four bacterial species studied) by following the tube dilution method. MBCs for the ATCC reference strains were also obtained using the same method. MBCs of none of the tested strains were recorded as being higher than the corresponding ATCC reference strains. The results for the MBC determination are displayed in Tables 5 and 6.

For 2% glutaraldehyde and 1.5% cresol, none of the MDR isolates showed growth at any of the dilutions tested (1:2 to 1:8). For 70% ethanol growth was observed for both the Gram-positive cocci and their corresponding ATCC reference strains at higher dilutions (1:4 and 1:8).

3.2. Detection of biocide resistance genes among MDR bacterial isolates

Biocide resistance genes were detected by performing PCR using *nfxB*, *qacG*, *smr*, and *qacA/B*-specific primers. Amplification was obtained using the *nfxB* primers for 87% of the *P. aeruginosa* isolates where a single amplification band was obtained in the 458 bp region. This finding was also replicated by performing in-silico PCR simulations (Fig. 1-Supplementary file). The identity of the amplification product was confirmed by sequencing to be the *nfxB* gene (matched with 100% identity and 100% coverage to NC_002516 GenBank reference sequence) (Figs. 2 and 3 - Supplementary file). No amplification band specific to the *nfxB* region was obtained for any of the other bacterial species studied. In the case of the *qacG* gene, amplification was detected in the 122 bp region for 6.67% (2 isolates) of *P. aeruginosa* isolates, which was also subsequently confirmed by

Table 3
Primer sequences used for the detection of biocide resistance genes.

Primers	Primer sequence 5'-3'	Amplicon length (bp)	Reference
<i>qacA/B</i> F (Gram-positive)	GCAGAAAGTGCAGAGTTTCG	361	Wu et al. [13]
<i>qacA/B</i> R	CCAGTCCAATCATGCCTG		
<i>smr</i> (<i>qacC</i> + <i>qacD</i>) F (Gram-positive)	GCCATAAGTACTGAAGTTATTGGA	195	Wu et al. [13]
<i>smr</i> (<i>qacC</i> + <i>qacD</i>) R	GACTACGGTTGTTAAGACTAAACCT		
<i>qacG</i> F (Gram-negative)	TCGCCTACGCAGTTTGGT	122	Wu et al. [13]
<i>qacG</i> R	AACGCCGCTGATAATGAA		
<i>nfxB</i> F (Gram-negative)	TGACCCTGATTCCCATGACG	194	Murugan et al. [14]
<i>nfxB</i> R	GTAGACCAGGGTGATGAACAGT		

Abbreviations: F – Forward primer, R – Reverse primer.

Table 4

PCR reaction conditions used for amplification.

Reaction step:	Reaction conditions:							
	<i>nfxB</i> primer set:			<i>qacG</i> primer set:			<i>qacA/B</i> and <i>smr</i> primer sets:	
Initial Denaturation	94 °C	5 min	×35 cycles	95 °C	2 min	×30 cycles	95 °C	5 min
Denaturation	94 °C	30 s		95 °C	25 s		95 °C	1 min
Annealing	55 °C	30 s		56 °C	25 s		58 °C	1 min
Extension	72 °C	90 s		72 °C	1 min		72 °C	1 min
Final extension	72 °C	7 min		72 °C	5 min		72 °C	5 min

Table 5

Growth in the presence of dilutions of 1% cetrimide (for MBC).

Organisms tested (Gram-negative bacilli):	Number of bacterial isolates showing growth at the biocide concentrations used (v/v %):		
	0.1% (100 mg/L)	0.01% (10 mg/L)	0.001% (1 mg/L)
<i>E.coli</i>	–	5	18
<i>K.pneumoniae</i>	2	9	20
<i>A.baumannii</i>	4	10	25
<i>P.aeruginosa</i>	6	12	23

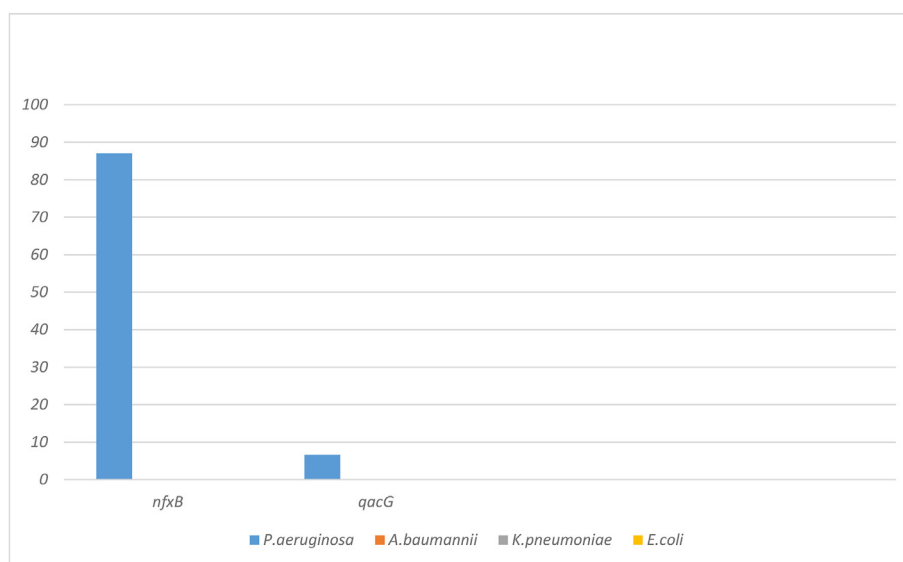
Table 6

Growth in the presence of dilutions of 1% Chlorhexidine (for MBC).

Organisms tested (Gram negative bacilli):	Number of bacterial isolates showing growth at the biocide concentrations used (v/v %):		
	0.1% (100 mg/L)	0.01% (10 mg/L)	0.001% (1 mg/L)
<i>E.coli</i>	–	5	18
<i>K.pneumoniae</i>	2	9	20
<i>A.baumannii</i>	4	10	25
<i>P.aeruginosa</i>	6	12	23

sequencing (Fig. 4- Supplementary file). The *qacG* genetic element was not detected in any of the other Gram-negative bacilli. The frequencies of detection of various biocide resistance genes among the Gram-negative bacterial isolates are displayed in Fig. 5. No genes for biocide resistance

(*qacA/B*, and *smr*) were detected among *S. aureus* and *E. faecium* isolates. However, it is interesting to note that the *qacA/B* and *smr* genes were frequently detected among Coagulase Negative Staphylococci such as *S. haemolyticus* and occasionally among *S. saprophyticus* (results not shown).

**Fig. 5.** Prevalence of biocide resistance genes among Gram negative bacterial isolates.

4. Discussion

This study was undertaken to address the burden of biocide resistance among MDR bacterial pathogens, which is an under-explored area in healthcare settings.

4.1. MBC determination by tube dilution method

The MBCs of the cationic biocides chlorhexidine and cetrimide tested against the Gram-negative organisms ranged from ≤ 0.75 to ≥ 150 $\mu\text{g/ml}$. Among the *E. coli* isolates, the lowest MBC recorded was ≤ 0.75 $\mu\text{g/ml}$ (for chlorhexidine), comparable to those obtained for *K. pneumoniae*, *A. baumannii* and *P. aeruginosa* isolates. However, the highest MBC value for the above biocides against *E. coli* isolates were 75 and 150 $\mu\text{g/ml}$ respectively, unlike the other three species, where a few isolates displayed MBCs of ≥ 75 and ≥ 150 $\mu\text{g/ml}$. Our findings were similar to those of Mazzola et al. [11]. No growth of test isolates was detected in any of the 2% glutaraldehyde and 1.5% cresol dilutions. However, it was noted that these biocides resulted in precipitation of the growth medium (Cation Adjusted Mueller Hinton Broth); hence, a modified protocol was used. Briefly, the tube dilution had to be performed in Trypticase Soy Broth (TSB) as described by Mazzola et al. along with residual biocide neutralization by subculture onto Soy Casein Digest Agar [11]. However, this process did not alter our initial observations. The test was replicated with CA-MHB after neutralization to cross-check the findings of the modification using the TSB, again the results remained unaltered.

Interestingly, none of the isolates showed MBC for any of the biocides higher than that of ATCC reference strains. This finding was in direct contrast to that reported by Gupta et al. in 2018, where a much higher proportion of the MDR bacterial isolates (of the same species) displayed resistance to some of the biocides tested in our study [2]. An earlier study demonstrated that MIC values for four different biocides tested against *E. coli* isolated from both environmental and clinical samples were comparable, and none of the isolates had a MIC value above the recommended concentrations of these biocides tested; an observation in agreement with ours [15]. A similar study by Bhatia et al. also demonstrated similar findings as ours [16]. This suggests that all MDR isolates need not be resistant to biocides unless proven otherwise.

Another salient point of note was that some of the Gram-positive cocci (both clinical isolates and corresponding reference strains) could withstand the actions of ethanol at lower concentrations (i.e. 35% and below) which might be attributable to their rigid Gram-positive cell wall rather than specific acquired resistance mechanisms. This finding also indicates that Gram-positive cocci may be inherently less susceptible to the actions of ethanol and may not be adequately taken care of by concentrations below the recommended level.

Subsequently, the MBC for the biocides was tested at their recommended concentrations using higher inoculum of the test organisms. Selected MDR isolates of *K. pneumoniae* and *E. coli* were challenged at increased inoculum (1 McF and 2 McF) with the disinfectants, 1% cetrimide, and 1% chlorhexidine. Cetrimide and chlorhexidine were chosen for this purpose as they belong to the class of quaternary ammonium compounds which are considered to be low-level disinfectants, and thus are biocides whose actions are more liable to be affected by variations in microbial load. For MDR *K. pneumoniae* whose MBC was recorded as 0.1% at standard inoculum (0.5 McF), a higher inoculum (i.e. 2 McF) resulted in growth being detected even at 1% cetrimide and 1% chlorhexidine concentration. This indicates, that in the case of higher-than-expected microbial load, the biocides (especially the low-level disinfectants such as quaternary ammonium compounds) in the recommended concentrations, may fail to achieve the purpose of disinfection.

4.2. PCR findings

In our study, biocide resistance genes were only detected in *P. aeruginosa* among all the bacterial species studied. The *nfxB* gene was

detected more frequently in *P. aeruginosa* compared to the *qacG* gene. The *nfxB* gene is a part of a master regulator for the Resistance-Nodulation-Division (RND) efflux pump *mexCD-OprJ*, which exports a broad range of substances including biocides such as chlorhexidine, organic solvents, and also antimicrobials such as fluoroquinolones [17,18]. Our isolates of *P. aeruginosa* which harbored the *nfxB* gene were resistant to many antibiotics including the fluoroquinolones. Hence, the presence and expression of such genes could result in cross-resistance to antibiotics and biocides.

The *qacG* genetic element was only detected in 6.67% of *P. aeruginosa* isolates a finding in accordance with its purported rarity reported in literature [13,19]. The *qacG* genetic element is a member of the Small Multidrug Resistance (SMR) family of transporters (which in Gram-positives also includes the *qacC*, *qacG*, *qacJ* and *qacH*) [20]. The *qacG2* gene in gram-negative bacterial isolates is frequently present as a part of the *-blaIMP-4-qacG2-aacA4-catB3* genetic cassette array which may lead to co-resistance to most beta-lactams except ceftazidime and piperacillin-tazobactam (via *blaIMP-4* carbapenemase), some aminoglycosides (via *aac*) and chloramphenicol (via *catB3*) [21]. This cassette array has been found in mobile genetic elements including plasmids of diverse Gram-negative bacterial species which can have dangerous implications for the widespread dissemination of biocide-antimicrobial co-resistance [22]. All isolates of *P. aeruginosa* included in this study were resistant to ceftazidime, piperacillin-tazobactam, meropenem, amikacin, and ciprofloxacin. Other *qac* classes such as *qacA/B* belong to the Major Facilitator Superfamily (MFS) and mediate resistance to a wide range of quaternary ammonium compounds and intercalating dyes (such as ethidium bromide and acriflavine) along with sharing significant homology with transporters for tetracycline in *S. aureus* [23,24].

Also, none of the isolates of *S. aureus* and *E. faecium* showed the presence of *qacA/B* and *smr*, despite these genes being detected among Coagulase Negative Staphylococci such as *S. haemolyticus* and *S. saprophyticus*. This finding probably suggests the presence of an environmental reservoir for these biocide-resistance genes, though this needs to be verified.

In our study, there was no significant correlation between the presence or absence of biocide-resistance genes and MBCs for the biocides in *Pseudomonas aeruginosa*. These findings are similar to those reported for MDR *K. pneumoniae*, *P. aeruginosa*, and *A. baumannii* by Vijayakumar et al. [25] and for *P. aeruginosa* by Pottier et al. [26].

5. Conclusions

Our findings suggest that all MDR isolates need not be resistant to biocides unless proven otherwise. Also, in the case of higher-than-expected microbial load, certain biocides (such as quaternary ammonium compounds) in the recommended concentrations may fail to achieve the purpose of disinfection. A few of the limitations of our study were that any association between any antibacterial resistance and specific biocide resistance was not explored; and although biocide resistance genes such as *nfxB* and *qacG* were detected, their expression patterns were not determined.

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No fund was received to conduct the study.

Conflict of interest

None.

Ethical approval with letter number

Approval for the study was obtained from the Institute Ethics committee (JIP/IEC/2021/263, dated August 19, 2021).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmm.2023.100478>.

References

- [1] Maillard JY. Antimicrobial biocides in the healthcare environment: efficacy, usage, policies, and perceived problems. *Therapeut Clin Risk Manag* 2005;1(4):307.
- [2] Gupta P, Bhatia M, Gupta P, Omar BJ. Emerging biocide resistance among multidrug-resistant bacteria: myth or reality? A pilot study. *J Pharm BioAllied Sci* 2018;10(2):96.
- [3] Russell AD. Introduction of biocides into clinical practice and the impact on antibiotic-resistant bacteria. *J Appl Microbiol* 2002;92:121S–35S.
- [4] Chuanchuen R, Beinlich K, Hoang TT, Becher A, Karkhoff-Schweizer RR, Schweizer HP. Cross-resistance between triclosan and antibiotics in *Pseudomonas aeruginosa* is mediated by multidrug efflux pumps: exposure of a susceptible mutant strain to triclosan selects *nfxB* mutants overexpressing MexCD-OprJ. *Antimicrob Agents Chemother* 2001;45(2):428–32.
- [5] Lambert RJ. Comparative analysis of antibiotic and antimicrobial biocide susceptibility data in clinical isolates of methicillin-sensitive *Staphylococcus aureus*, methicillin-resistant *Staphylococcus aureus* and *Pseudomonas aeruginosa* between 1989 and 2000. *J Appl Microbiol* 2004;97(4):699–711.
- [6] Fox LJ, Kelly PP, Humphreys GJ, Waigh TA, Lu JR, McBain AJ. Assessing the risk of resistance to cationic biocides incorporating realism-based and biophysical approaches. *J Industr Microbiol Biotech* 2022;49(1):kuab074.
- [7] Levy SB. Active efflux, a common mechanism for biocide and antibiotic resistance. *J Appl Microbiol* 2002;92:65S–71S.
- [8] Amsalu A, Sapula SA, De Barros Lopes M, Hart BJ, Nguyen AH, Drigo B, et al. Efflux pump-driven antibiotic and biocide cross-resistance in *Pseudomonas aeruginosa* isolated from different ecological niches: a case study in the development of multidrug resistance in environmental hotspots. *Microorganisms* 2020;8(11):1647.
- [9] Poole K. Mechanisms of bacterial biocide and antibiotic resistance. *J Appl Microbiol* 2002;92:55S–64S.
- [10] Meade E, Slattery MA, Garvey M. Biocidal resistance in clinically relevant microbial species: a major public health risk. *Pathogens* 2021;10(5):598.
- [11] Mazzola PG, Jozala AF, Novaes LC, Moriel P, Penna TC. Minimal inhibitory concentration (MIC) determination of disinfectant and/or sterilizing agents. *Braz J Pharm Sci* 2009;45(2):241–8.
- [12] Wales AD, Davies RH. Co-selection of resistance to antibiotics, biocides and heavy metals, and its relevance to foodborne pathogens. *Antibiotics* 2015;4(4):567–604.
- [13] Wu G, Yang Q, Long M, Guo L, Li B, Meng Y, et al. Evaluation of agar dilution and broth microdilution methods to determine the disinfectant susceptibility. *The J antibiotics* 2015;68(11):661–5.
- [14] Murugan N, Malathi J, Therese KL, Madhavan HN. Application of six multiplex PCR's among 200 clinical isolates of *Pseudomonas aeruginosa* for the detection of 20 drug resistance encoding genes. *Kaohsiung J Med Sci* 2018;34(2):79–88.
- [15] Ghanem B, Haddadin RN. Multiple drug resistance and biocide resistance in *Escherichia coli* environmental isolates from hospital and household settings. *Antimicrob Resist Infect Control* 2018;7(1):1–7.
- [16] Bhatia M, Loomba PS, Mishra B, Dogra V, Thakur A. Reduced susceptibility of carbapenem-resistant *Klebsiella pneumoniae* to biocides: an emerging threat. *Indian J Med Microbiol* 2016;34(3):355–8.
- [17] Fraud S, Campigotto AJ, Chen Z, Poole K. MexCD-OprJ multidrug efflux system of *Pseudomonas aeruginosa*: involvement in chlorhexidine resistance and induction by membrane-damaging agents dependent upon the AlgU stress response sigma factor. *Antimicrob Agents Chemother* 2008;52(12):4478–82.
- [18] Li XZ, Zhang L, Poole K. Role of the multidrug efflux systems of *Pseudomonas aeruginosa* in organic solvent tolerance. *J Bacteriol* 1998;180(11):2987–91.
- [19] Namaki M, Habibzadeh S, Vaez H, Arzanlou M, Safarirad S, Bazghandi SA, et al. Prevalence of resistance genes to biocides in antibiotic-resistant *Pseudomonas aeruginosa* clinical isolates. *Mol Biol Rep* 2022;49(3):2149–55.
- [20] Wassenaar TM, Ussery DW, Ingmer H. The *qacC* gene has recently spread between rolling circle plasmids of *Staphylococcus*, indicative of a novel gene transfer mechanism. *Front Microbiol* 2016;7:1528.
- [21] Espedido BA, Partridge SR, Iredell JR. *bla*-IMP-4 in different genetic contexts in Enterobacteriaceae isolates from Australia. *Antimicrob Agents Chemother* 2008;52(8):2984–7.
- [22] Kizny Gordon A, Phan HT, Lipworth SI, Cheong E, Gottlieb T, George S, et al. Genomic dynamics of species and mobile genetic elements in a prolonged *bla*-IMP-4-associated carbapenemase outbreak in an Australian hospital. *J Antimicrob Chemother* 2020;75(4):873–82.
- [23] Paulsen IT, Brown MH, Littlejohn TG, Mitchell BA, Skurray RA. Multidrug resistance proteins *QacA* and *QacB* from *Staphylococcus aureus*: membrane topology and identification of residues involved in substrate specificity. *Proc Natl Acad Sci USA* 1996;93(8):3630–5.
- [24] Dashtbani-Roozbehani A, Brown MH. Efflux pump mediated antimicrobial resistance by *Staphylococci* in health-related environments: challenges and the quest for inhibition. *Antibiotics* 2021;10(12):1502.
- [25] Vijayakumar R, Sandle T, Al-Aboody MS, AlFonaison MK, Alturaiki W, Mickymaray S, et al. Distribution of biocide resistant genes and biocides susceptibility in multidrug-resistant *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *Acinetobacter baumannii*—a first report from the Kingdom of Saudi Arabia. *J Infect Pub Health* 2018;11(6):812–6.
- [26] Pottier M, Gravey F, Castagnet S, Auzou M, Langlois B, Guerin F, et al. A ten-year microbiological study of *Pseudomonas aeruginosa* strains revealed diffusion of carbapenems and quaternary ammonium compounds resistant populations. *medRxiv [Preprint]*; 2022. <https://doi.org/10.1101/2022.11.06.22282008> [cited 2022 Dec 13]. [Accessed 13 December 2022].